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CS-499 Computer Science Capstone 2025 C-5 (Sep - Oct)

3-2 Milestone Two Enhancement One Software Design and Engineering

The artifact I selected for this enhancement is a travel web application built with Angular (frontend), Node.js/Express (backend), and MongoDB (database). It was originally created during CS-465: Full Stack Development. I chose to include this artifact in my ePortfolio because it highlights my ability to design, implement, and test a functional full-stack application.

This artifact demonstrates my skills in software design and engineering by showcasing how I can extend an existing system with new functionality. For this milestone, I enhanced the artifact by creating a RESTful API route for adding and removing favorites and implemented JWT-based authentication to ensure only authorized users can modify favorites. On the frontend I updated the Angular trip-card component to support dynamic UI updates based on user favorites. These enhancements demonstrate my abilities to plan improvements, apply best practices in software engineering, and deliver maintainable, scalable solutions.

In module one, I planned to design and implement solutions using algorithmic principles by adding a favorite feature, I believe I achieved these outcomes and applied a security mindset by requiring JWT authentication for all favorite actions, protecting user data, also demonstrated innovation by seamlessly extending both the backend and frontend while maintaining system stability.

Enhancing this artifact taught me how to extend a full stack application in a way that remains secure and reliable. I learned how to dynamically update Angular components and integrate new functionality without disrupting existing features.

One challenge I faced was handling TypeScript types for user IDs and trip objects, I also had to carefully integrate authentication into these new features to avoid breaking existing ones. By overcoming these challenges, I strengthened my skills in designing solutions that are both functional and secure. I also gained some experience in problem solving by checking if everything works accordingly.